Tendayi Kamucheka
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Summary

Compiler engineer working on ML and dataflow compilers for PIM/CIM architectures. My research focuses on the memory wall in AI acceleration: reducing data movement through graph partitioning, kernel selection, operator fusion, operand residency, and bus-traffic optimization. I designed and implemented an MLIR/LLVM compiler stack with custom dialects, code generation, runtime and driver integration, profiling, latency/throughput evaluation, and FPGA validation. I also bring experience with C++, Python, Linux, hardware–software co-design, and startup software development.

Technical Focus

- **Compiler & Memory Optimization**: MLIR, LLVM, ONNX lowering, graph partitioning, kernel selection, custom dialects/IRs, operator fusion, operand residency, bus-traffic minimization, memory-traffic analysis, lowering passes, code generation.
- **Performance & Runtime Systems**: Profiling, latency/throughput experiments, performance characterization, C++, Python, Linux, MicroBlaze V, ARM Cortex-A53, Vitis, bare-metal runtimes, accelerator-driver integration, simulation, systems debugging.
- **Accelerators & Co-Design**: PIM/in-memory computing, 2-D SIMD arrays, FPGA targets, hardware–software co-design, resource-tradeoff analysis, AXI Stream data movement, Verilog, custom ISA, custom assembly language.

Education

- **University of Arkansas** — Fayetteville, AR
  Doctor of Philosophy in Computer Engineering — *Jan. 2019 – Present (ABD)*
- **University of Arkansas** — Fayetteville, AR
  Master of Science in Computer Engineering — *Jan. 2019 – July 2026*
- **Nanjing University of Aeronautics & Astronautics** — Nanjing, China
  Bachelor of Engineering in Software Engineering — *Sept. 2012 – July. 2016*

Selected Projects

- **ML Compiler and Data-Movement Optimization**
  Graph partitioning, kernel selection, fusion, residency, and target code generation — *C++, Python, MLIR, LLVM, ONNX, ANTLR4, Verilog*
  - Designed and implemented a heterogeneous MLIR/LLVM compiler that performs graph partitioning and kernel selection, then lowers ONNX models through tensor-, memref-, and PIM-level dialects to programmable 2-D SIMD and PIM accelerator targets.
  - Developed compiler passes to maximize on-device operand residency and minimize bus traffic, including tile-residency analysis, static off-chip memory-traffic analysis, liveness-based register allocation, constant-load hoisting, and store/load forwarding.
  - Developed and evaluated operator-fusion and memory-placement passes through profiling, latency, and throughput experiments, using measured behavior to guide data-movement optimization across heterogeneous targets.
  - Implemented lowering for matrix multiplication, tiled reductions, transpose, elementwise operations, and activation functions on hardware without matching primitives.

Selected Projects — Continued

- **Compiler Runtime Integration and Portability**
  Cross-target execution, bare-metal deployment, FPGA implementation, and portability — *C++, Python, Vitis, Linux*
  - Designed a custom RISC-like ISA, assembly language, parser, encoder/decoder, and compiler import/export paths; aligned LLVM object and static-library generation with target ISA/ABI requirements and vendor build tools.
  - Retargeted the compiler across three accelerator architectures, MicroBlaze V (RISC-V) and ARM Cortex-A53 hosts, and eight synthesized processing-element array geometries using machine-readable platform descriptors.
  - Executed model workloads through Vitis on bare-metal MicroBlaze V and ARM Cortex-A53 FPGA systems and a custom PIM simulator; programmed Arty A7-100T, AUP-ZU3EG, AC701, VC707, ZCU104, and U55 platforms.
  - Owned synthesis, implementation, timing analysis/closure, runtime and accelerator-driver integration, and vendor-platform build compatibility from generated program to FPGA execution.
- **RTL and Validation Infrastructure**
  Cross-layer correctness, RTL/data-movement validation, and SoC integration — *Verilog, VHDL, Python, AXI Stream*
  - Validated compiler correctness with an eight-model neural-network suite against a CPU inference runtime and bit-exact comparisons across multiple array geometries.
  - Built functional simulation and directed RTL tests that localize failures across compiler semantics, register/layout assumptions, instruction sequencing, activations, busy/ready behavior, and hardware control.
  - Implemented RTL clock-domain-crossing logic and DMA read/write FIFO buffering for accelerator data movement, SoC integration, and FPGA realization.

Experience

- **University of Arkansas** — Fayetteville, AR
  Senior Graduate Assistant — *Jan. 2019 – Present*
  - **Research Assistant**: Developed ML and dataflow compilers with graph partitioning, kernel selection, operator fusion, profiling, latency/throughput evaluation, and data-movement optimization; integrated compiler output with FPGA RTL, runtimes, drivers, and C++/CUDA workloads.
  - **Teaching & Mentoring**: Designed embedded-systems lab coursework, taught computer organization and database management systems, and mentored graduate students in FPGA/RTL design and research workflows.
- **University of Southern California – Information Sciences Institute** — Arlington, VA
  Visiting Research Assistant — *May 2022 – August 2022*
  - **FPGA Validation**: Designed vendor-independent Verilog/VHDL slice-level tests; implemented parameterized slice targeting and custom LUT-to-FF routing under placement/routing constraints.
- **Purple Lite Media** — Nanjing, China
  Founder, Software Engineer — *September 2016 – March 2018*
  - **Software Engineering**: Founded a software startup, managed 3 developers and 2 content creators, and served as principal developer/architect for customer-facing information systems.
  - **Delivery & SDLC**: Led architecture decisions, coding and GitHub-based development workflows, CI/CD-oriented deployment, requirements and milestone planning, task assignment, and production delivery for client systems.

Publications

- Kabir, M. A., Fredricks, N., Kamucheka, T., Mandebi, J., Huang, M., Bakos, J. D., & Andrews, D. (2025). *DA-VinCi: A Deep-Learning Accelerator Overlay Using In-Memory Computing.* ACM Transactions on Reconfigurable Technology and Systems, 18(4), 1–38.
- Kamucheka, T., & Andrews, D. (2024, May). *Ph.D. Project: A Compiler-Driven Approach to HW/SW Co-Design of Deep-Learning Accelerators.* In 2024 IEEE 32nd Annual International Symposium on Field-Programmable Custom Computing Machines (FCCM) (pp. 237–238). IEEE.
- Kabir, M. A., Kamucheka, T., Fredricks, N., Mandebi, J., Bakos, J., Huang, M., & Andrews, D. (2024, September). *IMAGine: An In-Memory Accelerated GEMV Engine Overlay.* In 2024 34th International Conference on Field-Programmable Logic and Applications (FPL) (pp. 220–226). IEEE.
- Kabir, M. A., Kamucheka, T., Fredricks, N., Mandebi, J., Bakos, J., Huang, M., & Andrews, D. (2024, May). *The BRAM is the Limit: Shattering Myths, Shaping Standards, and Building Scalable PIM Accelerators.* In 2024 IEEE 32nd Annual International Symposium on Field-Programmable Custom Computing Machines (FCCM) (pp. 223–223). IEEE.